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Targeted modification of the carbon paste electrode by natural zeolite and graphene oxide for the enhanced analysis of paracetamol

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ABSTRACT

In this work, the targeted modification of carbon paste (CP) by admixing of conductive graphene oxide (GO) and mesoporous natural zeolite with the introduced Ni^{2+} cations (NiZ), providing electrocatalytic properties, is presented. The performed univariate optimization indicated that 5 % (w/w) of NiZ and GO in the CP is sufficient to ensure a threefold increase in peak current and lower the overpotential of paracetamol (PAR) oxidation, while retaining a low background current and high signal-to-noise ratio. The fabricated CPE/NiZ/GO sensor ensured a linear response in the PAR concentration range from 0.004 to 0.12 mg L^{-1}, advantageously low limit of detection equal to 0.0012 mg L^{-1} (0.0078 μM), and high stability of recorded signals (RSD = 3 %). Moreover, the proposed analytical method is characterized by good selectivity and robustness to the presence of interference. The usefulness of the CPE/NiZ/GO was verified by the determination of PAR in pharmaceuticals, human urine and serum, delivering acceptable results.

1. Introduction

Since its introduction to the electroanalysis [1], the Carbon Paste Electrode (CPE) has enjoyed remarkable interest among analytical chemists in the analysis of miscellaneous inorganic and organic depolarizers [2]. This can be attributed to its numerous advantages, including the ease of preparation by simple mixing of graphite with a water-immiscible liquid [3], a wide range of available potential [4], low electrical resistance [5], and high reproducibility [2]. In comparison to the Glassy Carbon Electrode (GCE) and noble-metal electrodes, CPE is a low-cost sensor with a low background current [6] and an easily restorable surface [7]. However, CPE has the propensity to poor resolution in multicomponent mixtures and slow transport rate [8]. To overcome these drawbacks, electrochemists have reported on numerous modifications of the carbon paste, including the use of graphene nanoribbons [9], graphene oxide (GO) [10], multi-walled carbon nanotubes (MWCNTs) [11], zeolites [5,12], TiO_2 nanoparticles [4,7], polymers such as Nafion [9] or polypyrrole [13], and carbohydrates [20]. The advantage of the carbon paste is the possibility of both the surface and bulk modification by deposition of a layer on the electrode surface and by adding a material directly into the carbon paste, respectively. The carefully chosen modifiers allow obtaining of a sensor with predetermined and desired features, like the amplified response towards a given analyte, lower detection limit, and higher sensitivity and selectivity [6,14]. Recently, the combination of highly conductive substances with materials displaying electrocatalytic properties in the fabrication of hybrid electrochemical sensors represents an important field of research in electroanalysis [10].

Zeolites are naturally occurring or synthetic aluminosilicates with a three-dimensional framework of channels and cages, made by the SiO_4 and AlO_4 tetrahedrons interconnected by oxygen bridges. This framework possesses a negative charge, resulting from the replacement of Si^{4+} by Al^{3+}, therefore, charge-compensating cations need to be introduced. When the transition metal cations are used for this purpose, the zeolite gains electrocatalytic properties [15]. Other advantages include the simplicity of functionalization, high thermal and mechanical stability,

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resistance to acids, and insolubility in most organic solvents. Zeolite-modified electrodes are characterized by increased sensitivity, low detection limits, selectivity, and reproducibility [16]. By dint of the heretofore mentioned advantages, zeolites have found use in the electroanalysis [17,18]. Despite synthetic zeolites being purer, the use of natural zeolites in sensors fabrication has become more common, because they are environmentally friendly and consistent with the Green Analytical Chemistry Guidelines [12].

Since its establishment, nanotechnology has been extensively applied in countless fields of state-of-the-art industry [19]. Due to their high electric conductivity and high surface area, carbon-based nanomaterials, including single walled carbon nanotubes (SWCNTs) [20] and MWCNTs [13], graphene nanoribbons [9], and graphene oxide (GO) [6,10], have been intensively employed as a modifier of the electrochemical sensors. GO offers numerous possibilities of functionalization by attaching oxygen-riches functional groups or by reduction and has a high specific surface area, enhancing its catalytic action [6]. The combination of materials of higher particle size with nanomaterials, known as nanocomposite, is the source of interesting properties from the point of view of electroanalysis.

Paracetamol (PAR) is one of the most-common over-the-counter, non-steroid medications used for reducing backache, headache, toothache, migraine, post-operative pain, and fever caused by bacterial or viral infections [8,16]. It constitutes an alternative to aspirin and phenacetin [19], as in safe dosage (max. 4 g per day [21]) it has little toxicity and only a few side effects [10]. Up to 10 % of the administered amount of PAR is excreted in the unchanged form in urine, the remaining portion is metabolized in the liver by glucuronidation (ca. 65 %), sulfation (ca. 25 %), and oxidation by cytochrome P450 isoenzyme. The latter leads to the formation of *N*-acetyl-*p*-benzoquinone imine (NAPQI) (ca. 10 %) [22], responsible for the toxic effect of PAR. It depletes the amount of glutathione [8] and has the ability to bind with proteins and liver cells [23,24], resulting in renal failure [2], gastritis, inflammation of the pancreas [19], and severe liver disease [23]. In the case of children, the administration of PAR increases the risk of asthma, allergies, and eczema [8,25]. Because of its easy availability, PAR has been involved in the considerable number of intentional drug overdose and attempted suicide cases [26].

These disadvantages justify the need for the development and improvement of the existing methods for PAR determination, in order for them to meet the requirements of the incumbent regulations regarding the determination of the PAR in biological fluids [12]. The standard methods used for this purpose include HPLC [27], HPLC with tandem mass spectrometry [28], UV–vis spectroscopy [29], and spectrofluorimetry [24]. Unfortunately, they are time-consuming, require sophisticated apparatus, and some of them are not suitable for the detection of low PAR concentrations [2]. Contradistinctively, electrochemical methods provide high accuracy, simplicity, and sensitivity of measurements [2], are relatively low-cost [16], and do not involve laborious sample preparation. Due to these inherent advantages, electrochemical methods have been applied in industrial and clinical analyses [30]. Regarding the electroanalysis of PAR, the usage of bare metallic and carbon-based electrodes has been reported [24,30]. To improve their performance, various modifications by means of metal- and metal-oxide nanoparticles [4,19], MWCNTs [3], and polymers [30] have been proposed.

In this work, the functionalization of the carbon paste with natural zeolite with the introduced Ni^{2+} ions, providing the electrocatalytic properties, and conductive graphene oxide (CPE/NiZ/GO) towards an enhanced voltammetric determination of paracetamol was presented. These materials have been carefully chosen based on their influence on PAR analytical signal and sensor performance. The phase composition and microstructure of the used natural zeolite have been investigated by means of powder diffraction, IR spectroscopy, and SEM. Furthermore, the mechanism of PAR oxidation at the electrode surface has been elucidated. The proposed procedure for PAR determination has been successfully employed for the analysis of PAR in pharmaceuticals, urine, and blood serum. The obtained values of relative errors, RSD, and recoveries proved the analytical usefulness of the proposed sensor and its outstanding stability. The main advantage and novelty of this work lie in the usage of relatively simple, easily-available, and stable materials, which ~~do not~~ require neither arduous preparation nor the addition of stabilizers. The proposed bulk modified carbon paste is exceptionally stable, so once prepared, it can be used in numerous experiments over a long time period, providing reproducible results.

2. Experimental

2.1. Chemicals and reagents

Graphite powder (<20 μm powder), paraffin oil, SWCNTs (>90 % carbon, 0.7–1.1 nm diameter), MWCNTs (>90 % carbon basis), Paracetamol standard (USP testing specifications), potassium hexacyanoferrate(II) trihydrate, potassium chloride (Suprapur), sodium nitrite, caffeine, Triton X-100, uric acid, glucose, sodium dodecyl sulfate (SDS), humic acids, and titanium dioxide were purchased from Sigma Aldrich. graphene (Industrial Quality) and graphene oxide (powder, S Method) were obtained from ACS Material, LLC. Cetrimonium bromide (CTAB), Na_2HPO_4, and NaH_2PO_4 were acquired from Avantor Performance Materials Poland S.A. All these reagents were of analytical grade and used without any further purification. The natural zeolite material from the Ukrainian-Transcarpathian Plant was received from the Andalusia Group (Poznań, Poland), whereas the synthetic zeolite Y (sodium form, CBV 100) was purchased from Zeolyst. The 1000 mg L^{-1} stock solution of PAR was prepared weekly by dissolving an appropriate amount of the standard in water and then was stored at 4 °C in the fridge. Working solutions of PAR at the concentration of 100 and 10 mg L^{-1} were prepared daily by diluting the stock solution with water. The phosphate buffer with a pH ranging from 6 to 8 was prepared by mixing the corresponding amounts of Na_2HPO_4, $NaHPO_4$, and water. For the preparation of all aqueous solutions, double-distilled water was used. Synthetic serum Nortrol (Thermo Scientific, Finland) and Seronorm Trace Elements Urine (Sero, Norway) were used to model the biological fluids.

2.2. Instrumentation

The phase composition of the used natural zeolite material was investigated using the X-ray powder diffraction X'Pert system (Philips, the Netherlands) with CuKα (λ = 1.5418 Å) radiation in the 2Θ angle range of 5–80° with a step of 0.016°. For qualitative analysis, X'Pert HighScore Plus software and International Centre for Diffraction Data (ICDD) were employed, whereas the quantitative phase composition was determined using the Rietveld method. To examine the structure of the zeolite framework, infrared spectra in the region of 4000 to 400 cm^{-1} (mid-infrared) were acquired with the use of Bruker Vertex 70v spectrometer (256 scans with the 2 cm^{-1} resolution). Prior to the measurement, the samples were mixed with KBr and subsequently pressed to a form of a pellet. The use of Tescan Vega3 Scanning Electron Microscope (ThermoFisher Scientific, USA) provided an insight into the morphology and structure of the zeolite before and after the ion-exchange with Ni^{2+}-ions, whereas the elemental mapping and EDS spectra, recorded by means of energy-dispersive X-ray spectroscopy EDS attachment, gave information about the chemical composition of these materials. The porosity of the zeolite material was established using the Barret-Joyner-Halenda (BJH) method for the interpretation of multipoint nitrogen adsorption data, recorded at 77 K (ASAP 2010, Micrometric Instruments, USA). Prior to the measurement, the sample was degassed for 24 h at 378 K.

All electrochemical measurements were conducted in the three-electrode cell consisting of the Pt wire auxiliary electrode, Ag|AgCl|3 M KCl reference electrode, and the carbon-paste-based working

electrode. Cyclic voltammograms (CV), differential pulse voltammograms (DPV), and impedance spectra were acquired using the potentiostat–galvanostat PGSTAT 128 N with the FRA32M impedance module (Metrohm Autolab B.V., Utrecht, the Netherlands). This apparatus was controlled by the Nova 2.1.4 software with the real-time acquisition feature. Between each measurement, the tested solution was stirred by a Teflon-coated bar. The pH measurements were performed with the use of the SevenCompact S210 laboratory pH-meter (Mettler Toledo, Switzerland).

2.3. Procedures

2.3.1. Preparation of the zeolite materials

To introduce the chosen cations into the zeolite framework, ion exchange has been performed according to the procedure described in our previous publications [17,18]. At first, the sodium cations of high affinity towards zeolite have been introduced into the zeolite framework with the aim to replace all naturally-present exchangeable cations. To that end, 1 g of zeolite was suspended in 4 mL of 2 M NaCl solution, shaken for 2 h, and then left for 24 h at room temperature to ensure the contact of zeolite with Na^+ cations. Subsequently, the zeolite was centrifuged at 15 000 rpm for 10 min., rinsed several times with a small portion of double-distilled water to remove the adsorbed Cl^- species [12] ($AgNO_3$ test), dried, and homogenized. Next, the received Na-form of zeolite was used to fabricate the Ni-form. For this purpose, the above-described procedure was adapted by changing the NaCl solution to 0.1 M $NiCl_2{\cdot}6H_2O$. To ensure the complete replacement of Na^+ by Ni^{2+} cations, the ion-exchange process was conducted twice using a fresh solution of $NiCl_2{\cdot}6H_2O$. The efficiency of Ni^{2+}-incorporation was verified by SEM/EDS measurements. A similar procedure has been applied to obtain the Ni-form of the Na-form of the purchased synthetic zeolite Y.

2.3.2. Fabrication of the carbon paste and the working electrode

For the preparation of the unmodified carbon paste, graphite and paraffin oil (weight ratio of 70:30) were mixed in a mortar for 30 min to ensure its homogeneity. The modified carbon pastes were prepared similarly by mixing the paraffin oil, graphite, and modifier(s) in the desired weight ratio. In every case, the used modifier replaced some portion of graphite so that the amount of oil (binder) remained the same for all fabricated pastes. The given percentage of the modifier relates to the whole mass of the paste (dry constituents with paraffin oil). After mixing in a mortar, the paste was transferred to a glass vessel and stored in a fridge overnight before its first usage. Once ready, the paste was packed into the cavity of a holder containing a platinum wire, and the surface exposed to the solution was first smoothed with a spatula and then polished on a weighing paper. The fabricated electrode was rinsed with distilled water and placed in the measuring cell. When not used, the CPE was submerged in distilled water, and the carbon pastes were stored in the fridge.

2.3.3. Voltammetric and impedance measurements

Upon submerging into the tested solution, all carbon-paste electrodes (both modified and unmodified) were conditioned by recording five cyclic voltammograms in the potential range from 0 to + 1.2 V, using the scan rate $\nu = 100$ mV s^{-1}. This step has ensured the repeatable and stable operation of the working electrode.

Cyclic (CV) and differential pulse voltammetry (DPV) have been applied at the stage of optimization of the paste composition. Subsequently, the influence of each modifier on the performance of the fabricated sensors was established using the CV and electrochemical impedance spectroscopy (EIS) techniques and 1 mM solution of $K_4[Fe(CN)_6]$ (redox indicator) in 1 M KCl acting as the supporting electrolyte. Cyclic voltammograms (CVs) were recorded in the potential range from −0.4 to 0.9 V using $\nu = 10$ mV s^{-1}, whereas the impedance spectra were acquired using the sinusoidal signals of frequency ranging from 100 kHz to 25 mHz and 10 mV amplitude, superimposed on the formal potential of $K_4[Fe(CN)_6]$. Based on the CVs recorded for 1 mM $K_4[Fe(CN)_6]$ with different values of ν, the active surface of fabricated sensors was calculated. All these measurements were performed in the deoxygenized solution.

To investigate the mechanism of PAR electrooxidation, CV has been used. The optimization of pH of the supporting electrolyte, calibration of the method, interference study, and the analysis of PAR content in the real samples have been performed using differential pulse (DP) voltammetry since this technique offers higher sensitivity, better signal resolution, and symmetrical peak shape. Unless otherwise specified, DP voltammograms were recorded in 0.1 M phosphate buffer (pH 7.4) using the potential step $E_s = 4$ mV, pulse amplitude $dE = 40$ mV, and modulation time $t_m = 20$ ms. For data analysis, each measurement was repeated three times. The detailed conditions for voltammetric measurements are specified in the corresponding chapters and the figure captions.

2.3.4. Sample preparation and analysis

Commercially available pharmaceutical formulations containing PAR were purchased in the local drugstores. For every tablet-form drug, three tablets were weighed, crushed in the mortar into a fine, homogenous powder. An appropriate weight amount was transferred quantitatively into the 50 mL volumetric flask and diluted to the mark with double-distilled water. The obtained suspensions were then subjected to sonication (20 min.) and filtration using a syringe filter (MPEC membrane) to avoid introducing any solid particles to the measuring cell. Each filtrate was diluted and then transferred to the measuring cell containing 0.1 M phosphate buffer (pH 7.4). The determination of PAR was carried out according to the standard addition method, which included the recording of DP voltammograms for blank and after the addition of the sample and PAR standard solution. At each step, three consecutive voltammograms were acquired, averaged, and subjected to baseline correction. In the case of *Pedicetamol* syrup, the appropriate amount (considering its density) was transferred to a 50 mL volumetric flask to obtain the 2 mg L^{-1} solution of PAR. Further analysis was performed similarly as for tablet-form drugs.

The certified reference material (CRM) of human urine and serum did not contain any PAR. Therefore, to study the possibility to apply the presented method for PAR determination in these matrices, they were spiked with a known amount of PAR. Due to the high protein content in serum, the precipitation of proteins using trichloroacetic acid (TCA) was performed [31]. To this end, 500 μL of serum was transferred to the Eppendorf tube, to which TCA was added to obtain the final concentration of 2 %. This mixture was vortexed for 30 s and then centrifugated for 30 min. at 14 500 rpm. The supernatant was filtered using the syringe filter, and the obtained filtrate was subjected to voltammetric analysis for PAR content according to the standard addition method. The preparation and analysis of urine were performed in a similar manner with the omission of the protein precipitation step.

3. Results

3.1. Characterization of the developed sensors and used materials

To enhance the sensor response towards PAR, the functionalization of the pure carbon paste by the incorporation of highly conductive and electrocatalytic materials is reported in this work. Natural zeolite Clinoptilolite and synthetic zeolite Y, both before and after the ion exchange, were tested as a potential electrocatalyst, whereas various carbon-based nanomaterials were examined for their influence on electrode charge-transfer kinetics.

3.1.1. Optimization of the carbon paste composition

The optimization of the carbon paste composition was performed according to the univariate approach and involved the examination of

the impact of different materials and their percentage in the paste on sensor's performance towards PAR. At first, unmodified Clinoptilolite (Clin), Clin with introduced Na^{+} ions (Na-Clin) and Ni^{2+} ions (Ni-Clin), as well as synthetic zeolite Y and zeolite Y with introduced Ni^{2+} cations (Ni-Y) were added to the carbon paste at the weight ratio of 10 % (separately). As presented in Fig. 1A and 1B, zeolite Y differs significantly from the other materials, as its admixing resulted in an elevated background current and decreased potential of Oxygen Evolution Reaction (OER). These phenomena are not desired in electroanalysis, as the recorded signals for analytes would be barely distinguishable from the background (especially in the case of ultra-low concentrations), and the early onset of OER narrows the range of the useful potential. For these reasons, zeolite Y has been excluded from further examination. The catalytic action of zeolite in electroanalysis is confirmed by the decrease in the overpotential of PAR oxidation and amplification of the oxidation peak current. According to Fig. 1A, the most advantageous performance is observed for Ni-Clin, therefore, it has been selected for subsequent experiments.

In the next step, the carbon pastes with different percentage ratios of Ni-Clin were fabricated. As depicted in Fig. 1C, the greatest decrease in the overpotential of PAR oxidation has been obtained for 5 % Ni-Clin, for which the difference between the potential of the anodic and cathodic peaks is the lowest. These observations indicate that the addition of zeolite to the carbon paste improves the reversibility of the studied electrode reaction. For higher zeolite contents, the decrease in overpotential and peak current is getting smaller, and the reverse peak is more diffuse, which can be attributed to the low electric conductivity of zeolites (in general). That is as following: for the low content of zeolite, its catalytic action prevails, however, as the zeolite content increases, the catalytic properties are suppressed due to the decremented electrical conductivity of the zeolite-containing carbon paste, which influences the uncompensated *iR* potential drop. The peak width does not change significantly with the addition of Ni-Clin, indicating that the charge transfer at the active centers is not hindered. The DP voltammograms (Fig. 1D) comply with the CVs presented in Fig. 1C. Taking into account the above explanation, the weight percent equal to 5 % of Ni-Clin has been selected as optimal and used in the following measurements. The working electrode made of carbon paste containing 5 % of Ni-Clin is hereafter referred to as CPE/NiZ.

Next, the influence of the 10 % addition of GO, graphene, MWCNTs, and SWCNTs to the carbon paste (containing 5 % Ni-Clin) on the signal of PAR was investigated. For all these nanomaterials, the increase in the PAR peak current, better reversibility of the electrode reaction, and lowering of the overpotential of PAR electrooxidation were observed (Fig. 1E and 1F). In the case of MWCNTs, the background current remained almost unchanged when compared with the CPE/NiZ, whereas for the remaining nanomaterials, it raises significantly, which can be ascribed to their high specific surface. Both the cyclic (Fig. 1E), as well as the DP voltammograms (Fig. 1F), indicated that the best signal-to-noise ratio has been provided by the GO, therefore, it was chosen for further studies. Fig. 1G and 1H present the CVs and DP voltammograms, respectively, recorded on the electrode with the carbon paste containing 5 % of Ni-Clin and various amounts of GO. Based on them it can be concluded that the peak current of PAR oxidation increased with the GO content up to 15 %, and then it became lower. However, a higher amount of GO is equivalent to the elevated background current and thus suppressed signal-to-noise ratio. Considering the main goal of the described modification, which is to fabricate a sensor that would allow for the detection and determination of the ultralow concentration of PAR, the carbon paste with 5 % of GO was selected as the most suitable for this purpose. When 10 % of GO had been used, the background noise would be on the same level as the signals corresponding to nanomolar concentrations of PAR, thus it would be hard to distinguish between them. Thus, the optimal composition of the carbon paste is 5 % of Ni-

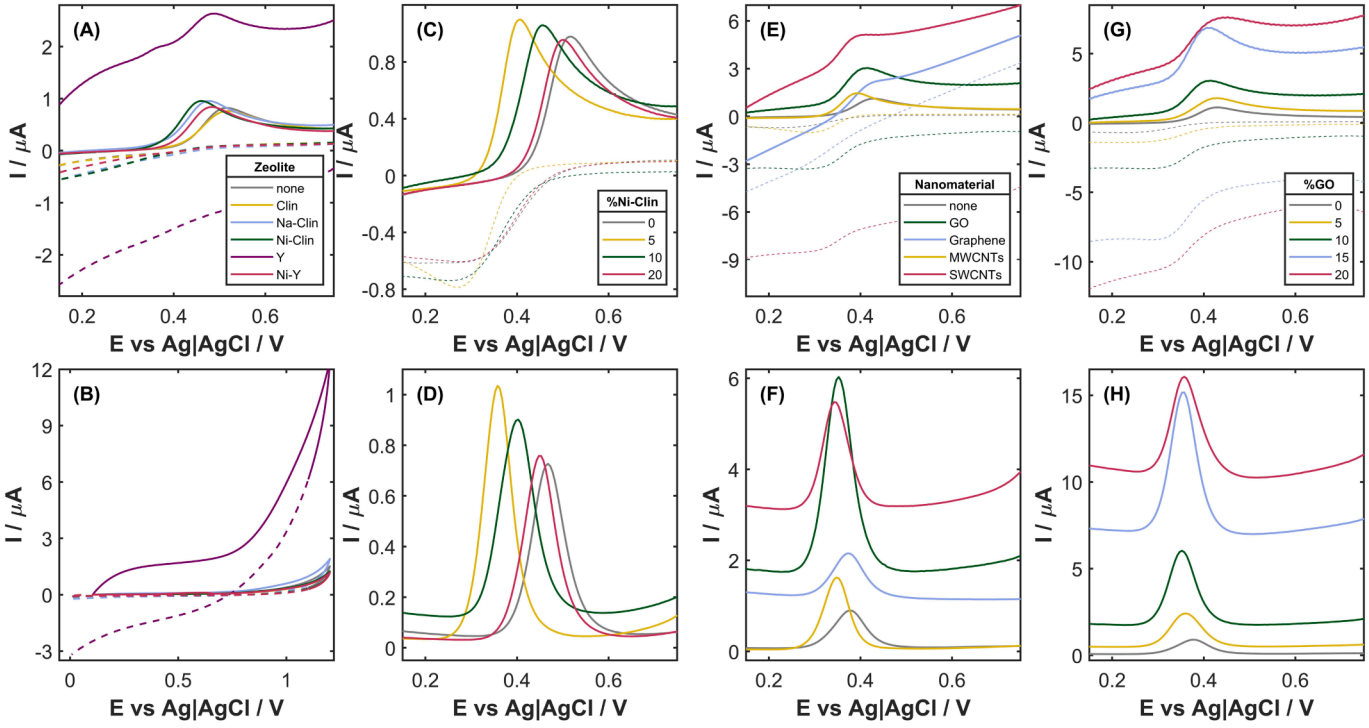


Fig. 1. Optimization of the carbon paste composition towards paracetamol (PAR) determination. (A) Cyclic Voltammograms (CVs) recorded for PAR and (B) pure phosphate buffer (pH 7.4) on the Carbon Paste Electrode (CPE) modified with different zeolite materials (10 % w/w). (C) CVs and (D) Differential Pulse Voltammograms (DPVs) recorded for PAR on the CPE with various percentages of Ni-Clinoptilolite. (E) CVs and (F) DPVs recorded for PAR on the CPE with 5 % Ni-Clinoptilolite (Ni-Clin) and 10 % of different (w/w) nanomaterial. (G) CVs and (H) DPVs for PAR on the CPE with 5 % Ni-Clin and various percentages of Graphene Oxide (GO). Concentration of PAR: (B) 0 mg L^{-1}, other figures – 2.5 mg L^{-1}. Supporting electrolyte: 0.1 M phosphate buffer (pH 7.4). Solid lines represent the anodic scans, whereas the dashed ones – the cathodic scans.

Clin, 5 % of GO, 60 % of graphite, and 30 % of paraffin oil. The electrode made of this paste is hereafter referred to as CPE/NiZ/GO.

The concept and significance of the low background current are better visualized in Fig. 2, in which CPE, CPE/NiZ, and CPE/NiZ/GO has been compared with the most-commonly used GCE in terms of the analytical signal they provided for 0.1 mg L^{-1} (0.67 μM) PAR. All tested electrodes gave a well-developed, symmetrical signal corresponding to PAR oxidation. However, in the case of GCE, the background current is elevated in comparison to the discussed CPEs. Further improvement is manifested in the curves obtained as a consequence of baseline correction, which are depicted as dashed lines in Fig. 2. At first, the potential of PAR oxidation has been significantly lowered for the CPE/NiZ/GO to + 0.350 V in comparison to CPE/NiZ (+0.365 V), CPE (+0.458 V), and GCE (+0.482 V). Moreover, the peak current provided by the CPE/NiZ/GO is 2 times higher than that of CPE/NiZ, 3 times higher than for CPE, and 4.1 times higher than for GCE, whereas the electrode surface area is equal ca. 5.8 mm^2 for carbon-paste based electrodes (see Section 3.1.3) and 7.1 mm^2 for GCE (MF-2012, φ = 3 mm, BASi, USA). This proves unequivocally that the improvement in sensor performance results from the used, carefully-chosen modifiers. Taking into account Fig. 2, non-arduous preparation, and ease in removal of any mechanical damage it can be concluded that the CPEs are more suitable for PAR determination than GCE is.

3.1.2. Structural and morphological characterization of used materials

Fig. 3A shows the XRD pattern of the zeolite material collected from the Ukrainian Sokirnyanskyy deposit and quantitative content of identified phases calculated by Rietveld's method. Interpretation of recorded diffraction data revealed that next to quartz and muscovite, the zeolite phase in the form of clinoptilolite constitutes the major component of the used material. Its quantitative content in the volcanic-sedimentary origin material equals 65.6 %. Clinoptilolite is classified to the heulandite group (HEU) and constitutes one of the most abundant zeolites among more than 40 known naturally occurring aluminosilicates. The theoretical composition of clinoptilolite is given by $(K_2,Na_2,Ca)_3[Al_6Si_{30}O_{72}]\cdot 24H_2O$ formula and its structure consists of secondary building units (SBU) built of 4-4 = 1 complex ($T_{10}O_{20}$ units). This complex is formed by 10 tetrahedra, which create two four-membered rings connected with an additional tetrahedron forming a five-membered ring (inset in Fig. 3B). Connection of these complexes in a three-dimensional framework results in the formation of an additional 8- and 10-membered ring of silico- and alumino-oxygen tetrahedra [32].

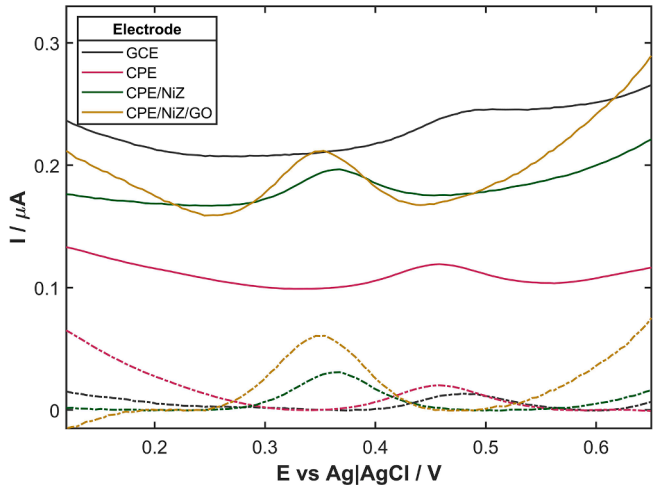


Fig. 2. Comparison of the performance of various working electrodes towards PAR determination. Solid lines depict the recorded DPVs, whereas the dashed curves represent the results of the baseline correction. Concentration of PAR equals to 0.1 mg L^{-1} (0.67 μM). Supporting electrolyte: 0.1 M phosphate buffer (pH 7.4).

Such building is responsible for the characteristic lamella structure of the HEU group of zeolites. The presence of 4-4 = 1 complex in clinoptilolite structure causes the appearance of the whole series of bands in the IR spectra (Fig. 3B), which can be divided into three groups according to the type of vibration realized in zeolite structure. The bands located in the range of 1200–400 cm^{-1} are related to internal Si-O(Si) and Si-O(Al) vibrations in SiO_4 and AlO_4 tetrahedra or alumino- and silico-oxygen bridges. The presence of water molecules in the zeolite structure results in bands in the range of 1600–3700 cm^{-1}. The most characteristic bands originating from over-tetrahedral structural units, i.e., rings built with a different number of members can be distinguished in the pseudo-lattice range of 800–500 cm^{-1} [33]. The band located at 674 cm^{-1} is most likely related to the vibration of two 4-membered rings, while the band at 607 cm^{-1} comes from the vibration typical for 5-membered rings [34].

The SEM images of the clinoptilolite before and after (Fig. 3C) the ion-exchange process has confirmed that the ion-exchange process has a negligible impact on the morphology and lamella structure of this material. This type of structure is characteristic for minerals from the clinoptilolite-heulandite family. Additionally, numerous voids can be observed in the SEM image of the clinoptilolite with introduced Ni^{2+} cations (Fig. 3C), which corroborates the porous nature. The average pore diameter, estimated based on the nitrogen adsorption isotherms (results not shown), was estimated to 18.82 nm, which allows classifying this material as mesoporous. The recorded elemental mappings (Fig. 3D) have shown the uniform distribution of Si, Al, O, which form the zeolite framework, and Ni in the sample. In the EDS spectrum of the unmodified zeolite material, clearly-distinguishable peaks corresponding to Si, Al, and O, were accompanied by smaller signals related to the presence of extra-framework cations, including K^+, Na^+, Mg^{2+}, and Ca^{2+}. For the zeolite after the ion-exchange process with Ni^{2+} cations, the height of the latter peaks diminished in favor of Ni^{2+} cations (Fig. 3E), proving the effectiveness of the conducted ion-exchange process. The atomic percentage of Ni^{2+} incorporated into the zeolite framework, estimated as the average from several surface point, is equal to 1.12 %.

3.1.3. Electrochemical properties of the fabricated sensors

To characterize the electrochemical properties of the developed sensors, CVs (Fig. 4A) and impedance spectra (Fig. 4B) were recorded for 1 mM $K_4[Fe(CN)_6]$ in 1 M KCl. Based on the CVs, the influence of the used modifier(s) on the reversibility of the electrode reaction, peak height, and electrode active surface was established. In the case of EIS measurement, the significant figures of merit were inferred by fitting the spectrum of an equivalent electrical circuit (EEC) to the recorded spectra. The EEC employed within this work is the commonly-known Randles cell, in which the capacitor has been substituted by the Constant Phase Element (CPE) (inset in Fig. 4B) to model the inhomogeneity and roughness of the electrode surface [35]. The determined parameters included the charge-transfer resistance (R_{ct}), Warburg coefficient (σ) for diffusion impedance, the differential capacity of the electrical double layer (C_{dl}), and the heterogeneous rate constant (k_s) (Table 1). When the impedance spectra are recorded at the formal potential of the used redox probe, the R_{ct} and σ are given by the following equations [36]:

$$R_{ct} = \frac{RT}{n^2F^2Ak_s}\cdot\frac{2\cdot(D_OD_R)^{1/4}}{D_O^{1/2}\cdot C_O^0 + D_R^{1/2}\cdot C_R^0} \quad (1)$$

$$\sigma = \frac{RT}{n^2F^2A\sqrt{2}}\cdot\frac{4}{D_O^{1/2}\cdot C_O^0 + D_R^{1/2}\cdot C_R^0} \quad (2)$$

where R stands for the universal gas constant ($R = 8.314$ J mol^{-1} K^{-1}), T for the temperature (in Kelvin), F for the Faraday constant (96485.3C mol^{-1}), A for the electrode active surface, n for the number of exchanged electrons per molecule, D_O and D_R for the diffusion coefficients of the oxidized and reduced forms, respectively (for $K_4[Fe(CN)_6]$ $D_{O/R} = 7.2\cdot 10^{-6}$ cm^2 s^{-1}), and C_O^0 and C_R^0 for their concentrations in the solution

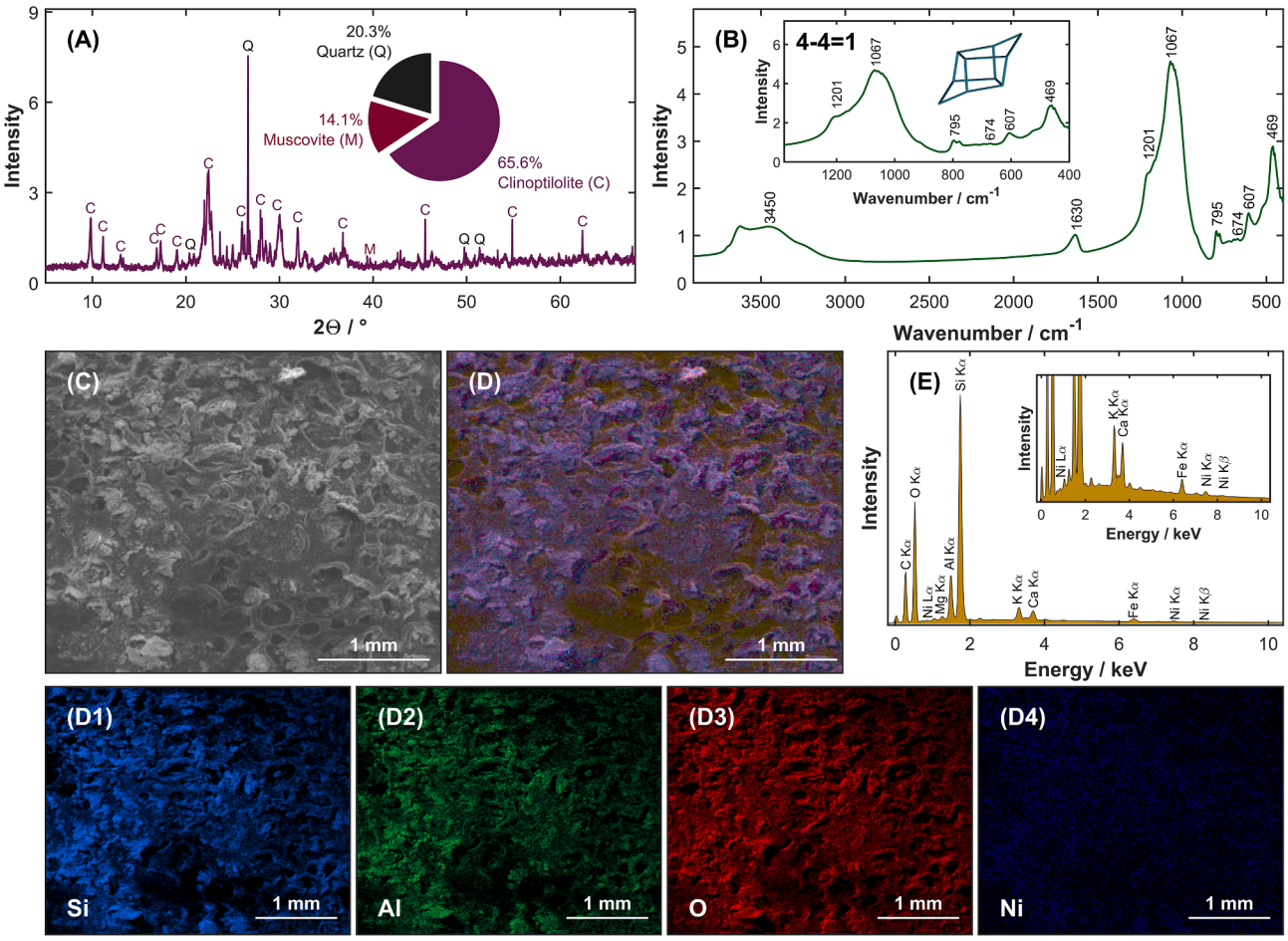


Fig. 3. (A) XRD pattern and (B) FT-IR spectrum of the used natural zeolite before the ion-exchange process. (C) SEM image of the zeolite after the ion-exchange with Ni^{2+} ions and the corresponding (D) layered elemental mapping, and (E) EDS spectrum (inset: the magnification of lower intensity values). (D1) Elemental mapping of Si, (D2) Al, (D3) O, and (D4) Ni.

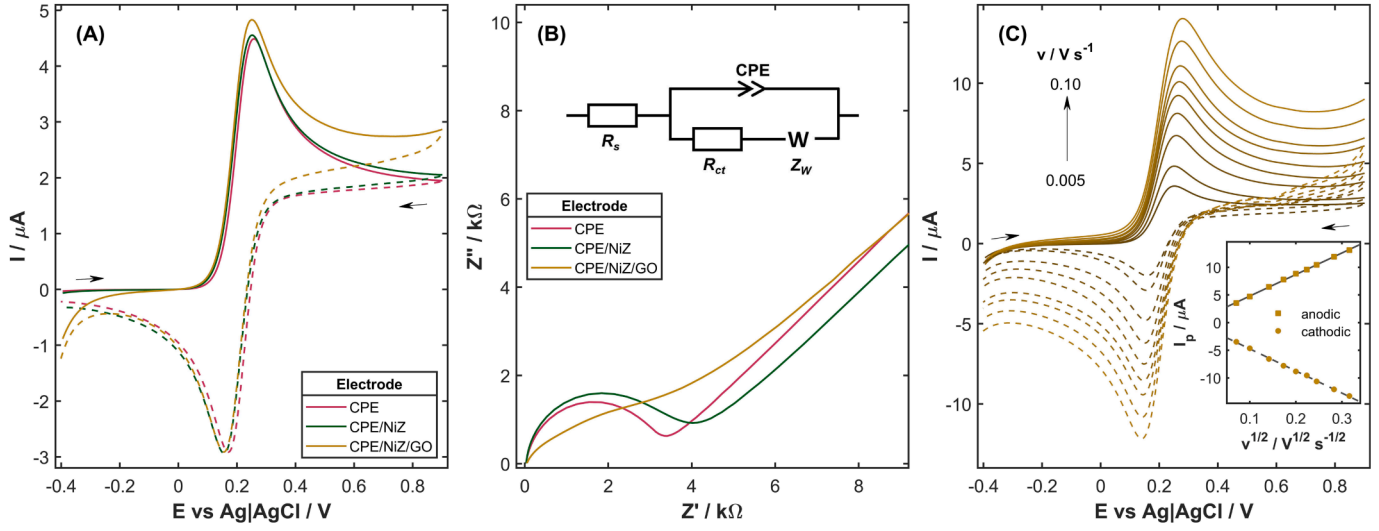


Fig. 4. (A) CVs recorded for 1.0 mM $K_4[Fe(CN)_6]$ in 1 M KCl on the unmodified Carbon Paste Electrode (CPE), CPE modified by the natural zeolite with the introduced Ni^{2+}-ions (CPE/NiZ), and CPE modified by the Ni-zeolite and Graphene Oxide (CPE/NiZ/GO). Scan rate $\nu = 10$ mV s^{-1}. (B) Corresponding impedance spectra (Nyquist plot). Inset: Randles cell equivalent circuit. (C) CVs recorded for 1 mM $K_4[Fe(CN)_6]$ in 1 M KCl on the CPE/NiZ/GO using various scan rates. Inset: the relationships between the peak current (I_p) and the square root of the scan rate (ν). In Fig. (A) and (C), the solid lines represent the anodic scan, and the dashed lines depict the subsequent cathodic scan.

Table 1
Electrochemical parameters of the tested electrodes calculated based on measurements for 1.0 mM $[Fe(CN)_6]^{4-}$ in 1 M KCl.

Parameter	Symbol	Unit	CPE	CPE/NiZ	CPE/NiZ/GO
Formal potential	E_f	mV	215 ± 2	205 ± 2	204 ± 2
Peak separation	ΔE	mV	89	95	94
Anodic peak current (v = 10 mV s^{-1})	I_{pa}	μA	4.47 ± 0.01	4.54 ± 0.01	4.73 ± 0.02
Anodic to cathodic peak current	I_{pa}/I_{pc}	–	0.97 ± 0.02	0.99 ± 0.02	1.00 ± 0.01
Electrode active surface	A	mm^2	5.8 ± 0.2	5.7 ± 0.2	5.9 ± 0.1
Charge transfer resistance	R_{ct}	kΩ	3.01 ± 0.09	3.5 ± 0.1	2.52 ± 0.04
Warburg coefficient	σ	kΩ $s^{-1/2}$	4.51 ± 0.03	4.80 ± 0.08	4.45 ± 0.05
Double layer capacitance	C_{dl}	μF cm^{-2}	2.06 ± 0.02	3.58 ± 0.03	14.3 ± 0.1
Heterogenous rate constant	k_s	m s^{-1}	(2.85 ± 0.05)·10^{-5}	(2.6 ± 0.1)·10^{-5}	(3.35 ± 0.08)·10^{-5}

(bulk concentrations). Division of equations (1) and (2) leads to the expression for k_s:

$$k_s = \frac{\sigma}{R_{ct}} \cdot \sqrt{\frac{D_{O/R}}{2}} \tag{3}$$

According to Fig. 4A and Table 1, the clinoptilolite with introduced Ni^{2+} cations (NiZ) and graphene oxide (GO) have caused only minor changes to the performance of the electrode towards ferrocyanide – the peak current increased only by 6 %, while for PAR, the peak current was tripled (compared to the bare CPE, see Fig. 2). For all tested electrodes, the obtained cyclic voltammograms have the shape characteristic for the reversible processes, as both the anodic and cathodic peaks of similar height are visible. The value of peak separation (ΔE, the difference between the anodic and cathodic peak potential) is elevated in comparison to the theoretical value, which can be ascribed to the uncompensated iR potential drop [37]. The impedance spectra (Fig. 4B), in which a semicircle, corresponding to the charge transfer, and a straight line, representing the diffusion of the redox probe, are visible, agree with the CVs. The R_{ct} has increased by 20 % upon the addition of NiZ, which results from its low electrical conductivity. This also means that NiZ evinces no catalytic activity towards $K_4[Fe(CN)_6]$ as opposed to PAR. The possible explanation for this phenomenon could be the same, negative charge of zeolite and $[Fe(CN)_6]^{4-}$ (the actual moiety, that is oxidized) and thus their mutual repulsion. Since at pH 7.4 PAR is a neutral molecule (pK = 9.7 [14]), no repulsion occurs. The addition of GO has lowered the R_{ct} by 17 % in comparison to the CPE, which is equivalent to the increase in the k_s value. The addition of both modifiers has increased the differential capacity (C_{dl}) of the electrical double layer, which can be observed in an elevated background current. The Warburg coefficient is nearly the same for the examined electrodes, indicating that their active surfaces are similar.

To calculate the electrode active surfaces of each electrode, CVs were recorded in 1 M solution of KCl containing 1 mM $K_4[Fe(CN)_6]$, using scan rate (v) values ranging from 0.005 to 0.1 V s^{-1}. Fig. 4C presents the results of this proceedings for CPE/NiZ/GO. As can be seen in the inset, for both the cathodic and anodic scans, the peak current is proportional to the square root of v. The surface A can be thus computed by comparing the slope of the $I_p(v^{1/2})$ dependency with the theoretical slope, predicted by the Randles-Ševčík equation [38]:

$$I_p = 0.4463\left(\frac{n^3F^3}{RT}\right)^{1/2} AD^{1/2}v^{1/2}C^0 \tag{4}$$

where I_p is the peak current, and v is the scan rate. The averaged values of A, calculated based on the anodic and cathodic scans, have been summarized in Table 1. They indicate that the modification influences the surface to a negligible extent, thus any change in peak current and position must be ascribed to the employed modifier.

3.2. Electroanalysis of paracetamol on the CPE/NiZ/GO

3.2.1. Investigation of the PAR oxidation mechanism

The electrochemical behavior of PAR on the CPE/NiZ/GO has been investigated based on the influence of the scan rate v and pH of the supporting electrolyte on the position and height of the recorded peak. According to Fig. 5A, with the increase in v, the anodic peak current (I_{pa}) raises, the anodic peak potential (E_{pa}) shifts towards more positive values (especially for high v), and the reverse peak has become less pronounced. By plotting the $\ln(I_{pa})$ values vs $\ln(v)$ (Fig. 5B), the linear relationship with the slope of 0.543 ± 0.003 has been obtained. This value is characteristic for the processes controlled either by diffusion of the depolarizer or by charge-transfer kinetics [38], however, the above-mentioned changes in E_{pa} with v indicate that the second option is much more fitting. For the kinetics-controlled processes, peak potential varies with v according to the following formula:

$$E_p = K + \frac{RT}{2\alpha nF}\ln(v) \tag{5}$$

where K is a constant value, and α signifies the charge transfer coefficient. From the substitution of the physiochemical constants and T = 298 K (25 °C), the slope of the above relationship amounts to 0.0128/αn, whereas in the case of PAR, it is equal to 0.0143 (Fig. 5C). Assuming the value of α = 0.5 (typically adopted value [39]), the number of exchanged electrons n = 1.8, which indicates the participation of two electrons in PAR oxidation reaction.

In the next step, the impact of pH on the PAR oxidation peak was examined. For this purpose, DPVs for 2.5 mg L^{-1} PAR were recorded in 0.1 M phosphate buffer solutions of pH ranging from 6 to 8 (Fig. 6A). The received curves exhibit a shift towards less positive potential with the increase in pH value, which points to the proton exchange accompanying the electron transfer. In such case, the Nernst equation can be written as follows [40]:

$$E = E^o + \frac{RT}{nF}\ln\frac{[O]}{[R]} - (59\ mV)\cdot\frac{p}{n}\cdot pH \tag{6}$$

where p denotes the number of exchanged protons. It can be thus inferred that when the number of exchanged protons and electrons are the same, the slope of the linear relationship between peak potential and pH is equal to 59 mV per unit change in pH. In the studied case, the obtained E_p(pH) dependency fulfills these criteria (Fig. 6B), which means that two protons participate in PAR oxidation. Considering these findings and the potential at which the reaction occurs, it must proceed via the oxidation of two carbons in the benzene ring, leading to the formation of *N*-acetyl-*p*-benzoquinone imine (NAPQI) [14] – the same compound that is produced by the oxidation of PAR inside the human body during the oxidation of PAR by cytochrome P450. Additionally, Fig. 1A indicates that the observed catalytic effect is associated with the presence of Ni^{2+}-cations inside the zeolite framework, thus they have to participate in the electrode process. Based on these observations, a reaction mechanism has been proposed, which covers the exchange of ions between the zeolite (denoted as z), solution (s), and the electrode-solution interface (esi) [41] and takes into account the calculated number of transferred protons and electrons (Fig. 6C). It involves the formation of the complex between Ni^{2+}-ions from the zeolite with PAR molecule [42], which leads to the increased concentration of PAR in the electrode vicinity, and thus amplified current and reduced overpotential of the studied electrooxidation reaction. All DPVs presented in Fig. 6A have a symmetrical peak corresponding to PAR oxidation. Nevertheless, the highest peak current has been obtained in the phosphate buffer of pH 7.4, therefore, it has been selected as the best supporting electrolyte and

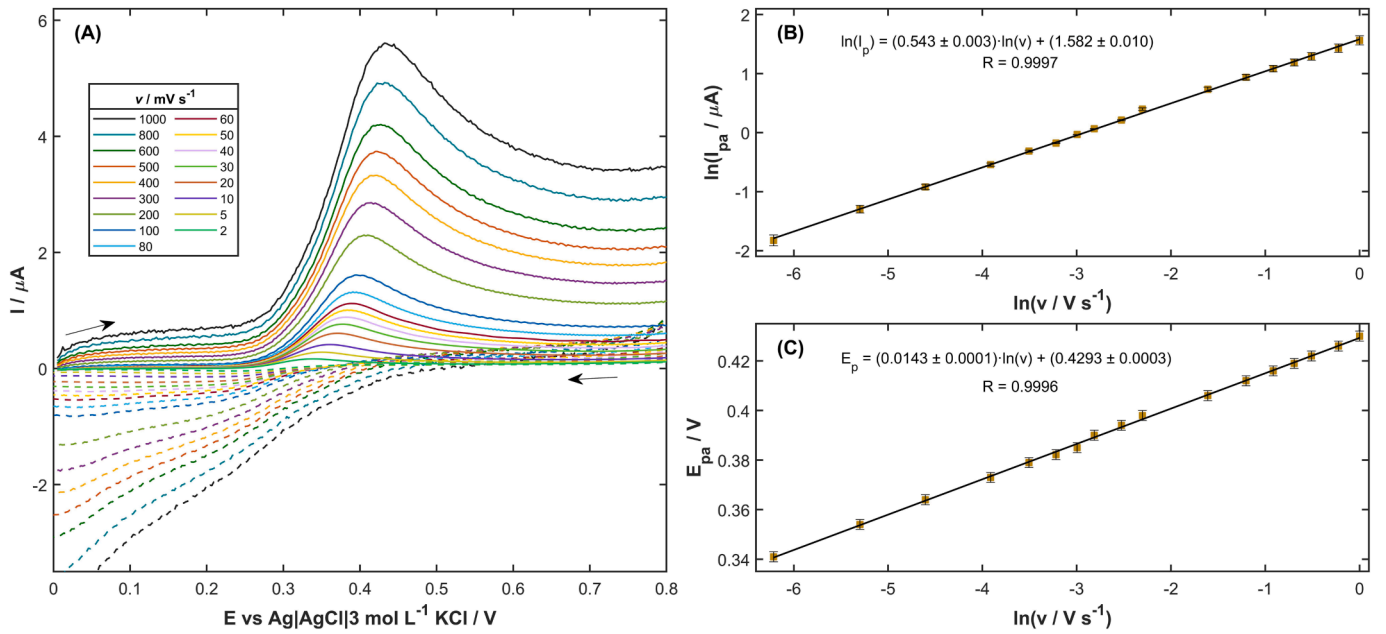


Fig. 5. Investigation of the mechanism of the electrode reaction. (A) CVs recorded for the increasing scan rate values v (given in the legend) for 2.5 mg L^{-1} (16.5 μM) PAR in 0.1 M phosphate buffer (pH 7.4). Solid lines represent the anodic scans, whereas the dashed lines stand for cathodic scans. (B) The relationship between the natural logarithm of the oxidation peak current (I_p) and the natural logarithm of the scan rate (v). (C) The dependency of oxidation peak potential (E_p) on the natural logarithm of v.

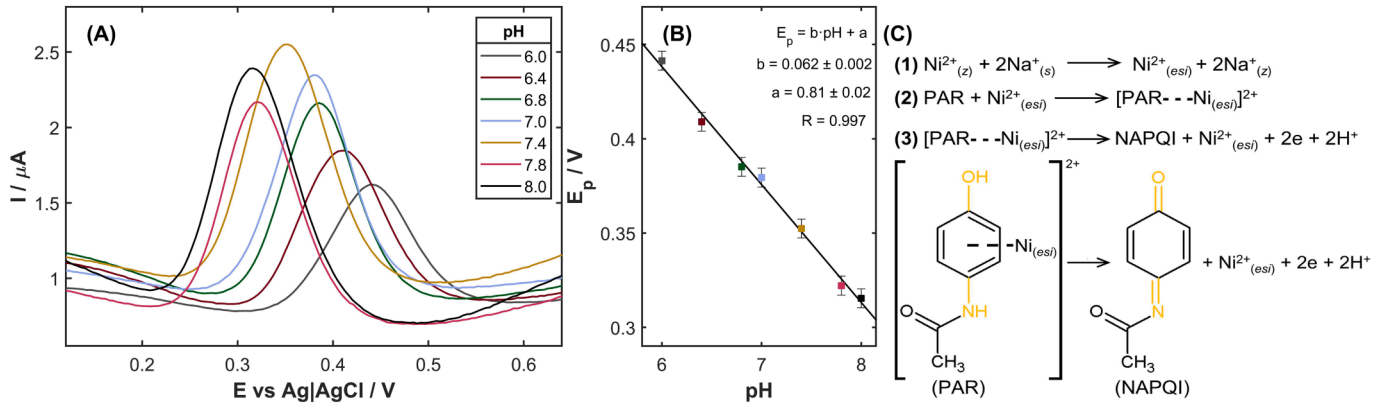


Fig. 6. (A) DPVs recorded for oxidation of 2.5 mg L^{-1} PAR in 0.1 M phosphate buffer of different pH values. (B) The relationship between the peak potential (E_p) and the pH of the supporting electrolyte. (C) Proposed reaction mechanism.

employed in the following experiments.

3.2.2. *Analytical performance, repeatability, and reproducibility of the sensor*

To validate the proposed method, DP voltammograms were recorded on the CPE/NiZ/GO sensor for the increasing concentrations of PAR. As presented in Fig. 7A, a well-defined, symmetrical peak of small half-width has been obtained. For every PAR addition, three consecutive scans were recorded, averaged, and subjected to the baseline correction using the 3rd-degree polynomial, leading to the curves marked in orange. Plotting the peak current values vs the PAR concentration (C_{PAR}) resulted in the linear relationship, depicted in Fig. 7B. Based on that, the figures of merit, such as range of linear response, sensitivity (slope), linearity, and limit of detection (LOD) and quantification (LOQ), were established, and the calculated values have been summarized in Table 2. LOD and LOQ were computed as the standard deviation of the peak current recorded for blank divided by the slope b and multiplied by 3.3 and 10, respectively [43]. This proceeding resulted in LOD of 0.0012 mg L^{-1} (0.0078 μM) and LOQ of 0.004 mg L^{-1} (0.026 μM).

The received parameters have been compared with the ones reported in the literature for the modified CPEs and summarized in Table 3. Based on that, it can be concluded that the protocol presented in this work is distinguished by a high sensitivity and wide linear range. The LOD at the nanomolar level is comparable or even much more favorable in comparison to the results described in the literature so far with the use of (bi) metal and metal-oxide nanoparticles [10,19,25], carbon nanomaterials [11,44], and organic polymers [30]. This offers the possibility of determining the trace level of PAR in a multiplicity of samples and enables their dilution to reduce the undesired influence from the matrix components. It is worth mentioning that the modifiers used in this work require neither any sophisticated apparatus nor laborious pretreatment. Quite the contrary, their preparation can be carried out using simple instrumentation, available in a chemical laboratory, and once conducted, these materials retain their stability over a long period, with no need of any stabilizer. Thus, once prepared, they can be applied in a considerable number of measurements, without any deterioration in properties. This has been confirmed by the RSD values of the peak height recorded for 0.1 mg L^{-1} PAR (three consecutive scans) on ten separately

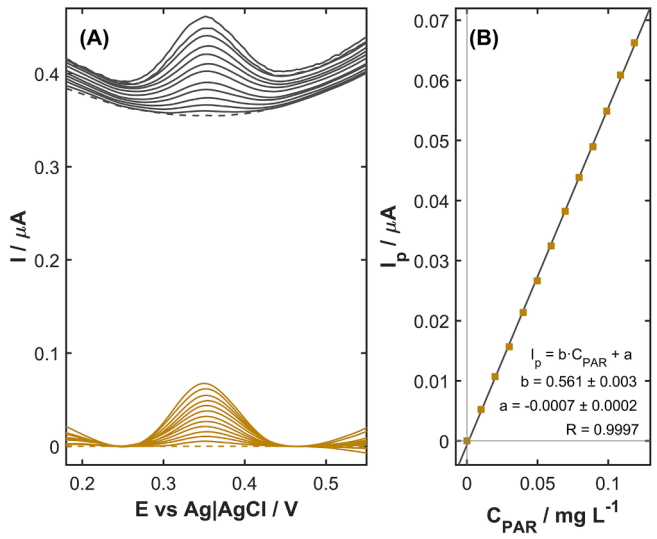


Fig. 7. (A) DPV measurements of the increasing concentration of PAR in the range from blank to 0.12 mg L^{-1} in the 0.1 M phosphate buffer (pH 7.4) using the CPE/NiZ/GO working electrode (gray curves) with the results of the baseline correction (marked in orange). (B) The resulting calibration plot with the regression equation.

Table 2
Figures of merit of the developed method for PAR determination with the usage of CPE/NiZ/GO.

Parameter	Unit	Value
Linear range	mg L^{-1}	0.004 – 0.12
Intercept *a*	μA	−0.0007 ± 0.0002
Slope *b*	μA L mg^{-1}	0.561 ± 0.003
R	–	0.9997
LOD	mg L^{-1} (μM)	0.0012 (0.0078)
LOQ	mg L^{-1} (μM)	0.004 (0.026)

prepared electrode copies. For each of them, RSD did not exceed 4.5 %, thus confirming the exceptional repeatability and short-time stability of every prepared electrode. The reproducibility, expressed as the RSD of the peak currents provided by all examined copies, amounted to 24 %, thereby corroborating the stability of the carbon paste composite and fabricated sensors.

3.2.3. Interference studies

In the next part of the research, the robustness of the described method to the presence of numerous chemical entities has been investigated. The tested chemicals included inorganic ions, organic compounds, and surface-active species, which are usually present in the analyzed pharmaceutical and biological samples. In the case of inorganic ions (Na^+, K^+, Ca^{2+}, Mg^{2+}, Cl^-, NO_3^-, and SO_4^{2-}), 0.1 M solution of a corresponding salt has been added to 0.1 M phosphate buffer (pH 7.4) containing 0.1 mg L^{-1} (0.67 μM) PAR in order to achieve 10-, 50-, 100-, 500-, and 1000-fold excess. Similarly, for glucose, caffeine, uric acid, Triton X-100 (non-ionic surfactant), cetrimonium bromide (CTAB, cationic surfactant), sodium dodecyl sulfate (SDS, anionic surfactant), humic acids, 0.1 % solution of each substance was separately added as to obtain the concentrations of interferents equal to 1.0, 5.0, 10, 25, 50, and 100 μM. In the case of titanium dioxide and starch, 5 mg of the powder was gradually (per 1 mg) added directly to the voltammetric cell. For every substance, the percentage difference (PD) between the peak current recorded for 0.1 mg L^{-1} PAR in the absence and in the presence of an interferent was calculated for every addition. In the case of PD less than ± 5 %, the given interferent was assumed to have little or no influence on the analytical signal.

In the course of the conducted measurements, no influence (PD < 1 %) of any tested inorganic ions on the position, height, shape, and baseline of PAR peak has been stated, even for the highest concentration. The same applies to the TiO_2, starch, and SDS. In the case of glucose and caffeine, the signal of PAR oxidation decreased by 4.5 %, however, its shape remained intact. With the subsequent additions of Triton X-100 and CTAB, the peak current was getting lower, reaching the 23 % and 32 % diminution, respectively, for the highest tested concentrations. Thereby, to comply with the above-mentioned tolerance limit, the concentration of Triton X-100 and CTAB should not exceed 25 and 50 μM, respectively. The suppressing effect is even more visible in the case of humic acids, for which the 50 % decrease in peak height and elevated background current was observed. Thus, the presence of surface-active species in the studied samples has to be considered, as it may pose difficulties during analysis and may require resorting to more advanced calibration approaches [43]. Another significant interferent is uric acid (UA), which gives rise to a well-defined oxidation peak at + 0.25 V, that is close to the oxidation peak of PAR. In the case of high excess of UA, these peaks overlap, thereby making the interpretation more difficult. Nevertheless, it can be easily resolved by an appropriate dilution of the sample containing UA, like urine.

Table 3
Characteristics of bulk-modified Carbon Paste Electrodes (CPEs) applied for PAR determination.

Electrode	Tech.	LOD	Linear range	Samples	Ref.
coffee husks/CPE[1]	SWV	0.66 μM	6.6–550 μM	tablets	[2]
MWCNTs/$ZnCrFeO_4$/CPE[2]	DPV	$1.4 \cdot 10^{-6}$ g/L	$1.5 \cdot 10^{-5}$-$5.6 \cdot 10^{-2}$ g/L	pharmaceutical and biological samples	[3]
Brilliant Blue-CPE[3]	CV	0.21 μM	5–30 μM	pharmaceuticals	[8]
Au-Pd/RGO/CPE[4]	DPV	$1.1 \cdot 10^{-6}$ g/L	$4.5 \cdot 10^{-6}$-$1.4 \cdot 10^{-3}$ g/L	urine, serum, pharmaceuticals	[10]
Poly(L-Dopa)/MWCNTs/CPE[5]	DPV	$3 \cdot 10^{-6}$ g/L	$9 \cdot 10^{-6}$ – $4.5 \cdot 10^{-3}$ g/L	serum	[11]
Fe(III)-Clino/CPE[6]	SWV	0.15 μg/L	$1.5 \cdot 10^{-6}$- 15 g/L	human serum, urine, tablets	[12]
Glucose-based CPE	DPV	$3.72 \cdot 10^{-8}$ M	0.6–9.0 μM	pharmaceuticals, urine	[14]
Co-ZA/CPE[7]	SWV	0.04 μM	0.1–190 μM	pharmaceuticals	[16]
MgO-NPs/CTAB/CPE[8] MgO-NPs/TX-100/CPE[9]	CV	6.2 μM	10–100 μM	pharmaceuticals	[19]
MoO_3-CPE	SWV	0.14 μM	0.36–6.0 μM	syrup and tablets	[25]
Chitosan-CPE	SWV	$5.08 \cdot 10^{-7}$ M	$2 \cdot 10^{-8}$-$8 \cdot 10^{-7}$ M	natural waters, tablets, urine	[30]
CNTPE[10]	DPV	$4.4 \cdot 10^{-5}$ g/L	$(2.3–27) \cdot 10^{-3}$ g/L	water, wastewater	[44]
ZnS-NPs @ Graphene – CPE[11]	DPV	6.5 nM	0.07–780 μM	pharmaceuticals, human blood plasma	[45]
CPE/NiZ/GO	DPV	0.0012 mg L^{-1} (0.0078 nM)	0.004 – 0.12 mg L^{-1}	pharmaceuticals, urine, serum	this work

[1] CPE modified with coffee husks from the coffee-processing mill; [2] CPE modified with multi-walled carbon nanotubes and magnetic $ZnCrFeO_4$ nanoparticles; [3] CPE with Brilliant Blue (organic triarylmethane dye); [4] CPE with Au-Pd bimetallic nanoparticles and Reduced Graphene Oxide; [5] CPE with complex of Levodopa Polymer and multi-walled carbon nanotubes; [6] CPE with nanoparticles of Fe(III)-clinoptilolite; [7] CPE with zeolite A with the introduced Co^{2+}-ions; [8] CPE containing MgO nanoparticles co-precipitated from the solution of MgO and CTAB (Cetrimonium bromide) or [9] TX-100 (Triton X-100); [10] Carbon Nanotubes Paste Electrode; [11] CPE modified with ZnS nanoparticles and graphene;

3.2.4. Practical applications of the sensor

In order to verify the applicability of the CPE/NiZ/GO, this sensor has been applied for PAR determination in pharmaceuticals and biological matrices (Certified Reference Materials) (Fig. 8). Each analysis was performed triply for statistical evaluation, which included the assessment of mean value, its standard deviation, and RSD. In the case of tablets, the relative error has been calculated based on the declaration of the manufacturer (Table 4). Since urine and serum matrices did not contain PAR, they were spiked with a known amount of PAR, and the recovery parameter has been calculated. The results are summarized in Table 5.

For all tested pharmaceuticals, a single peak was observed in the potential range corresponding to PAR oxidation and the height increased after the addition of PAR standard solution (Fig. 8A). Thus, as confirmed in course of interference studies, components of the pharmaceuticals do not distort the analytical signal to a degree, which would make the determination more difficult. The RE values below 5 % indicate satisfactory compliance with the content given by the manufacturer, and the RSD lower than 4.3 % proves a very good precision of the developed protocol. Based on that, it can be stated that the use of CPE/NiZ/GO for the determination of PAR content in pharmaceuticals will provide accurate and reliable results.

In the case of urine, an additional peak at + 0.25 V has been observed, which corresponds to the oxidation of UA (see Section 3.2.3). Nevertheless, the peak resulting from PAR oxidation can be clearly distinguished, and its height increased for the subsequent additions of PAR standard (Fig. 8B). These increments were, however, not as big as in pure phosphate buffer or in the case of pharmaceutical analysis (Fig. 8A), which was caused by the matrix components. This action is even more pronounced for highly complex serum samples (Fig. 8C). Fortunately, no additional peak was observed in the potential range corresponding to PAR oxidation. The recoveries received for spiked urine and serum oscillate around 100 %, proving the accuracy of the method. For urine, the RSD lies below 2.5 %, whereas in the case of the serum – below 9 %, which results from the small values of recorded current. Altogether, these results show that the described voltammetric method with CPE/NiZ/GO sensor can be successfully used in routine analyses of PAR.

Table 4
Determination of PAR in pharmaceuticals.

Sample	Declared value of PAR/ mg	Amount found/ mg	RE/ %	RSD/ %
Accord	500 / tablet	507 ± 9	1.5	1.8
Acenol	500 / tablet	477 ± 18	−4.6	3.9
Mexalen	500 / tablet	515 ± 5	3.0	1.0
Pedicetamol	100 / mL	101 ± 4	1.1	4.3

Table 5
Recovery test of PAR in spiked biological matrices.

Sample	Addition of PAR/mg L^{-1}	Amount found/mg L^{-1}	Recovery/ %	RSD/ %
Urine	30	29.2 ± 0.2	97.3	0.5
	60	61.3 ± 1.5	102.2	2.4
Serum	15	14.6 ± 1.3	97.2	8.9
	30	29.9 ± 1.9	99.8	6.3

4. Conclusions

Modern electrochemistry focuses on the search of new sensors and honing the existing ones. Carbon Paste Electrodes (CPEs) are commonly used due to the simplicity of their preparation and usage, ease of both the bulk and surface modification. By admixing the electrocatalytic and highly conductive materials to the carbon paste, the fabrication of the sensor with desired properties is possible. In this regard, mesoporous natural zeolite and graphene oxide were used within this article, and the obtained sensor (CPE/NiZ/GO) has provided the amplified response towards paracetamol (PAR), marked by the catalytic effect. This functioning is specific for PAR since only negligible improvement has been noted for the model redox system. The developed sensor has ensured a detection limit at the level of 0.0012 mg L^{-1} (0.0078 μM) and a wide range of linear response. Its applicability has been confirmed in the analysis of PAR as the main constituent in pharmaceuticals, as well as its trace amounts in biological samples, like human urine and serum.

CRediT authorship contribution statement

Radosław Porada: Conceptualization, Methodology, Investigation, Validation, Formal analysis, Data curation, Visualization, Writing – original draft, Writing – review & editing. **Nadine Wenninger:** Methodology, Investigation, Validation, Resources. **Christina Bernhart:** Methodology, Investigation, Validation, Resources. **Katarzyna Fendrych:** Investigation, Formal analysis, Resources, Writing – original

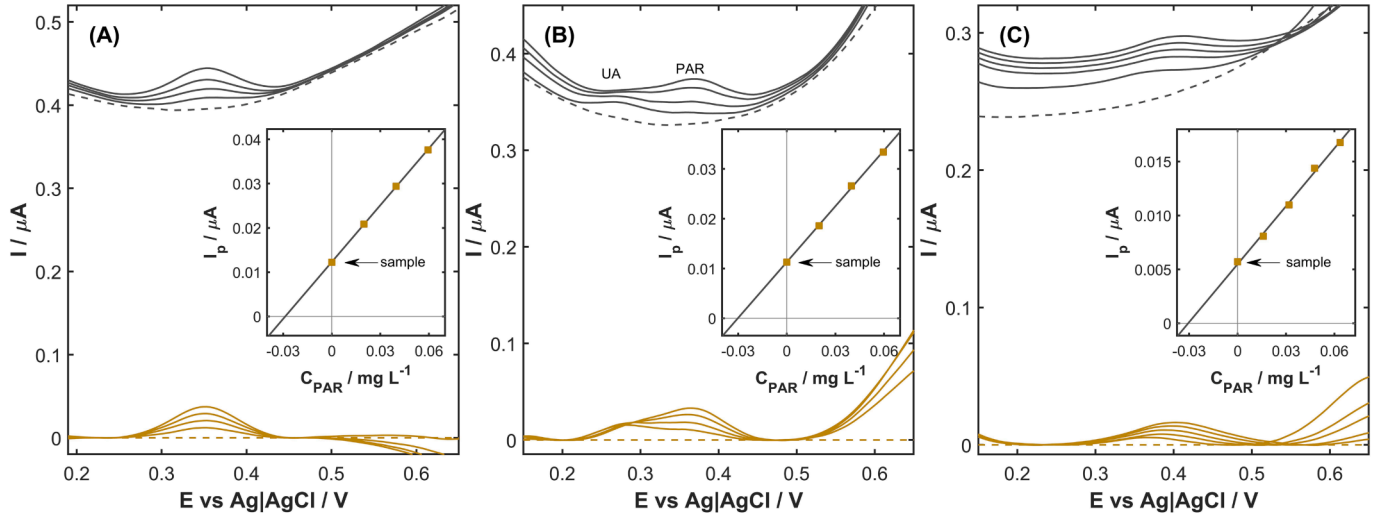


Fig. 8. Determination of PAR in (A) *Acenol* pharmaceutical, (B) spiked human urine, and (C) spiked human serum. Grey lines represent the DP voltammograms recorded for blank (dashed line), the addition of the sample, and subsequent additions of PAR standard solution, whereas the orange curves depict the result of baseline correction. Inset: corresponding standard addition calibration plot. UA – uric acid.

draft. **Jolanta Kochana:** Conceptualization, Supervision, Writing – review & editing. **Bogusław Baś:** Conceptualization, Methodology, Data curation, Supervision, Writing – review & editing. **Kurt Kalcher:** Supervision, Writing – review & editing. **Astrid Ortner:** Conceptualization, Methodology, Data curation, Supervision, Writing – review & editing.

Declaration of Competing Interest

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Data availability

Data will be made available on request.

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References

[1] R.N. Adams, Carbon paste electrodes, Anal. Chem. 30 (1958) 1576, https://doi.org/10.1021/ac60141a600.

[2] S.F. Mbokou, M. Pontié, J.-P. Bouchara, F.M.M. Tchieno, E. Njanja, A. Mogni, P. Y. Pontalier, I.K. Tonle, Electroanalytical performance of a carbon paste electrode modified by coffee husks for the quantification of acetaminophen in quality control of commercialized pharmaceutical tablets, Int. J. Electrochem. 2016 (2016) 1–10, https://doi.org/10.1155/2016/1953278.

[3] M. Taei, H. Salavati, F. Hasanpour, S. Habibollahi, H. Baghlani, Simultaneous determination of ascorbic acid, acetaminophen and codeine based on multi-walled carbon nanotubes modified with magnetic nanoparticles paste electrode, Mater. Sci. Eng. C 69 (2016) 1–11, https://doi.org/10.1016/j.msec.2016.06.014.

[4] M.H. Mashhadizadeh, F. Rasouli, Design of a new carbon paste electrode modified with TiO_2 nanoparticles to use in an electrochemical study of codeine and simultaneous determination of codeine and acetaminophen in human plasma serum samples, Electroanalysis 26 (2014) 2033–2042, https://doi.org/10.1002/elan.201400141.

[5] M. Abrishamkar, S.E. Tilami, S.H. Kaldozakh, A. Branch, K. Science, Electrocatalytic oxidation of cefixime at the surface of modified carbon paste electrode with synthesized nano zeolite, Adv. J. Chem. A 3 (2020) 767–776. 10.22034/AJCA.2020.114113.

[6] A.M. Fekrya, S.A. Abdel-Gawad, S.M. Azab, A. Walcarius, A sensitive electrochemical sensor for moxifloxacin hydrochloride based on nafion/graphene oxide/zeolite modified carbon paste electrode, Electroanalysis 33 (2021) 964–974, https://doi.org/10.1002/elan.202060355.

[7] J.B. Raoof, R. Ojani, M. Baghayeri, Simultaneous electrochemical determination of glutathione and tryptophan on a nano-TiO_2/ferrocene carboxylic acid modified carbon paste electrode, Sensors Actuators, B Chem. 143 (2009) 261–269, https://doi.org/10.1016/j.snb.2009.08.046.

[8] P.S. Ganesh, S.Y. Kim, S. Kaya, R. Salim, G. Shimoga, S.H. Lee, Quantum chemical studies and electrochemical investigations of polymerized brilliant blue-modified carbon paste electrode for in vitro sensing of pharmaceutical samples, Chemosensors 9 (2021), https://doi.org/10.3390/chemosensors9060135.

[9] N. Wenninger, U. Bračič, A. Kollau, K. Pungjunun, G. Leitinger, K. Kalcher, A. Ortner, Development of an electrochemical sensor for nitric oxide based on carbon paste electrode modified with Nafion, gold nanoparticles and graphene nanoribbons, Sensors Actuators B Chem. 346 (2021), https://doi.org/10.1016/j.snb.2021.130532.

[10] F. Tadayon, S. Vahed, H. Bagheri, Au-Pd/reduced graphene oxide composite as a new sensing layer for electrochemical determination of ascorbic acid, acetaminophen and tyrosine, Mater. Sci. Eng. C 68 (2016) 805–813, https://doi.org/10.1016/j.msec.2016.07.039.

[11] H. Yang, Z. Wei, S. He, T. Li, Y. Zhu, L. Duan, Y. Li, J. Wang, Fabrication of electrochemical sensor for acetaminophen based on levodopa polymer and multi-walled carbon nanotubes complex, Int. J. Electrochem. Sci. 12 (2017) 11089–11101, https://doi.org/10.20964/2017.12.50.

[12] S. Sharifian, A. Nezamzadeh-Ejhieh, Modification of carbon paste electrode with Fe (III)-clinoptilolite nano-particles for simultaneous voltammetric determination of acetaminophen and ascorbic acid, Mater. Sci. Eng. C. 58 (2016) 510–520, https://doi.org/10.1016/j.msec.2015.08.071.

[13] I. Zrinski, S. Martinez, A. Ortner, A. Samphao, J. Zavašnik, K. Kalcher, E. Mehmeti, A novel sensor based on carbon paste electrode modified with polypyrrole/multi-walled carbon nanotubes for the electrochemical detection of cytostatic drug rapamycin, Electroanalysis 33 (2021) 1325–1332, https://doi.org/10.1002/elan.202060527.

[14] V.B. Patil, R.R. Sawkar, D. Ilager, N.P. Shetti, S.M. Tuwar, T.M. Aminabhavi, Glucose-based carbon electrode for trace-level detection of acetaminophen, Electrochem. Sci. Adv. (2021) 1–9, https://doi.org/10.1002/elsa.202100117.

[15] A. Walcarius, Zeolite-modified electrodes in electroanalytical chemistry, Anal. Chim. Acta 384 (1999) 1–16, https://doi.org/10.1016/S0003-2670(98)00849-6.

[16] L. Ahmadpour-Mobarakeh, A. Nezamzadeh-Ejhieh, A zeolite modified carbon paste electrode as useful sensor for voltammetric determination of acetaminophen, Mater. Sci. Eng. C 49 (2015) 493–499, https://doi.org/10.1016/j.msec.2015.01.028.

[17] M. Madej, K. Fendrych, R. Porada, M. Flacha, J. Kochana, B. Baś, Application of Fe (III)-exchanged clinoptilolite/graphite nanocomposite for electrochemical sensing of amitriptyline, Microchem. J. 160 (2021), 105648, https://doi.org/10.1016/j.microc.2020.105648.

[18] R. Porada, K. Fendrych, B. Baś, Electrochemical sensor based on Ni-exchanged natural zeolite/carbon black hybrid nanocomposite for determination of vitamin B6, Microchim. Acta 188 (2021) 323, https://doi.org/10.1007/s00604-021-04992-x.

[19] K.G. Manjunatha, B.E.K. Swamy, H.D. Madhuchandra, K.J. Gururaj, K. A. Vishnumurthy, Synthesis and characterization of MgO nanoparticle and their surfactant modified carbon paste electrode sensing for paracetamol, Sensors Int. 2 (2021), 100127, https://doi.org/10.1016/j.sintl.2021.100127.

[20] K. Sipa, M. Brycht, A. Leniart, S. Skrzypek, The application of carbon nanomaterials as electrode surface modifiers for the voltammetric sensing of nitroxinil – a comparative study, J. Electroanal. Chem. 848 (2019), 113294, https://doi.org/10.1016/j.jelechem.2019.113294.

[21] M. Sakthivel, M. Sivakumar, S.M. Chen, Y.S. Hou, V. Veeramani, R. Madhu, N. Miyamoto, A facile synthesis of $Cd(OH)_2$-rGO nanocomposites for the practical electrochemical detection of acetaminophen, Electroanalysis 29 (2017) 280–286, https://doi.org/10.1002/elan.201600351.

[22] M.R. McGill, H. Jaeschke, Metabolism and disposition of acetaminophen: recent advances in relation to hepatotoxicity and diagnosis, Pharm. Res. 30 (2013) 2174–2187, https://doi.org/10.1007/s11095-013-1007-6.

[23] J.A. Hinson, D.W. Roberts, L.P. James, Mechanisms of Acetaminophen-Induced Liver Necrosis, in: U. J. (Ed.), Uetr. J. Advers. Drug React. Handb. Exp. Pharmacol. Vol. 196, Springer Berlin Heidelberg, 2010: pp. 369–405. 10.1007/978-3-642-00663-0.

[24] H. Montaseri, P.B.C. Forbes, Analytical techniques for the determination of acetaminophen: a review, TrAC – Trends Anal. Chem. 108 (2018) 122–134, https://doi.org/10.1016/j.trac.2018.08.023.

[25] E. Nagles, M. Ceroni, C. Villanueva Huerta, J.J. Hurtado, Simultaneous electrochemical determination of paracetamol and allura red in pharmaceutical doses and food using a Mo(VI) oxide-carbon paste microcomposite, Electroanalysis 33 (2021) 2335–2344, https://doi.org/10.1002/elan.202100261.

[26] C. Daly, E. Griffin, E. Mcmahon, P. Corcoran, R.T. Webb, D.M. Ashcroft, E. Arensman, Paracetamol – related intentional drug overdose among young people: a national registry study of characteristics, incidence and trends, 2007–2018, Soc. Psychiatry Psychiatr. Epidemiol. 56 (2021) 773–781, https://doi.org/10.1007/s00127-020-01981-y.

[27] B.S. Nagaralli, J. Seetharamappa, B.G. Gowda, M.B. Melwanki, Liquid chromatographic determination of ceterizine hydrochloride and paracetamol in human plasma and pharmaceutical formulations, J. Chromatogr. B 798 (2003) 49–54, https://doi.org/10.1016/j.jchromb.2003.08.048.

[28] H. Li, C. Zhang, J. Wang, Y. Jiang, J.P. Fawcett, J. Gu, Simultaneous quantitation of paracetamol, caffeine, pseudoephedrine, chlorpheniramine and cloperastine in human plasma by liquid chromatography-tandem mass spectrometry, J. Pharm. Biomed. Anal. 51 (2010) 716–722, https://doi.org/10.1016/j.jpba.2009.10.009.

[29] M.R. Khoshayand, H. Abdollahi, M. Shariatpanahi, A. Saadatfard, A. Mohammadi, Simultaneous spectrophotometric determination of paracetamol, ibuprofen and caffeine in pharmaceuticals by chemometric methods, Spectrochim. Acta – Part A 70 (2008) 491–499, https://doi.org/10.1016/j.saa.2007.07.033.

[30] Y. El Bouabi, A. Farahi, N. Labjar, S. El Hajjaji, M. Bakasse, M.A. El Mhammedi, Square wave voltammetric determination of paracetamol at chitosan modified carbon paste electrode: application in natural water samples, commercial tablets and human urines, Mater. Sci. Eng. C. 58 (2016) 70–77, https://doi.org/10.1016/j.msec.2015.08.014.

[31] J. Lu, E.L. Frank, Rapid HPLC measurement of thiamine and its phosphate esters in whole blood, Clin. Chem. 54 (2008) 901–906, https://doi.org/10.1373/clinchem.2007.099077.

[32] D.W. Breck, Zeolite Molecular Sieves: Structure, Chemistry, and Use, John Wiley & Sons, New York, 1973.

[33] W. Mozgawa, The influence of some heavy metals cations on the FTIR spectra of zeolites, J. Mol. Struct. 555 (2000) 299–304, https://doi.org/10.1016/S0022-2860(00)00613-X.

[34] W. Mozgawa, The relation between structure and vibrational spectra of natural zeolites, J. Mol. Struct. 596 (2001) 129–137, https://doi.org/10.1016/S0022-2860(01)00741-4.
[35] R. Porada, K. Fendrych, B. Baś, The Mn-zeolite/graphite modified glassy carbon electrode: fabrication, characterization and analytical applications, electroanalysis 32 (2020) 1208, https://doi.org/10.1002/elan.201900744.
[36] T. Zajt, Metody woltamperometryczne i elektrochemiczna spektroskopia impedancyjna, Wydawnictwo Gdańskie, Gdańsk, 2001.
[37] N. Elgrishi, K.J. Rountree, B.D. McCarthy, E.S. Rountree, T.T. Eisenhart, J. L. Dempsey, A practical beginner's guide to cyclic voltammetry, J. Chem. Educ. 95 (2018) 197–206, https://doi.org/10.1021/acs.jchemed.7b00361.
[38] S.E. Group, Instrumental Methods in Electrochemistry, Ellis Horwood Limited, Chichester, West Sussex, 1990.
[39] K. Sipa, M. Brycht, A. Leniart, P. Urbaniak, A. Nosal-Wiercińska, B. Pałecz, S. Skrzypek, β-Cyclodextrins incorporated multi-walled carbon nanotubes modified electrode for the voltammetric determination of the pesticide dichlorophen, Talanta 176 (2018) 625–634, https://doi.org/10.1016/j.talanta.2017.07.084.
[40] M.D. Lovander, J.D. Lyon, D.L. Parr, J. Wang, B. Parke, J. Leddy, Electrochemical properties of 13 vitamins: a critical review and assessment, J. Electrochem. Soc. 165 (2018) G18–G49, https://doi.org/10.1149/2.1471714jes.
[41] C. Senaratne, J. Zhang, M.D. Baker, C.A. Bessel, D.R. Rolison, Zeolite-modified electrodes: intra- versus extrazeolite electron transfer, J. Phys. Chem. 100 (1996) 5849–5862, https://doi.org/10.1021/jp951923x.
[42] I. Ledeti, G. Simu, G. Vlasp, T. Vlase, T. Olariu, G. Savoiu, L.M. Suta, C. Popow, A. Fulias, Ni(II) coordination compound with acetaminophen synthesis and characterization, Rev. Chim. 65 (2014) 556–559.
[43] M. Madej, J. Kochana, B. Baś, Determination of viloxazine by differential pulse voltammetry with boron-doped diamond electrode, Monatshefte Fur Chemie 150 (2019) 1655–1665, https://doi.org/10.1007/s00706-019-2380-6.
[44] F.A. Gorla, E.H. Duarte, E.R. Sartori, C.R.T. Tarley, Electrochemical study for the simultaneous determination of phenolic compounds and emerging pollutant using an electroanalytical sensing system based on carbon nanotubes/surfactant and multivariate approach in the optimization, Microchem. J. 124 (2016) 65–75, https://doi.org/10.1016/j.microc.2015.07.021.
[45] N. Soltani, N. Tavakkoli, F. Shahdost-fard, F. Davar, A. Kochakalipour Ranjbar, Applicability of ZnSNP@Gr nanocomposite for fabrication of an electrochemical sensor in simultaneous measuring of naltrexone, acetaminophen and ascorbic acid, Chem. Pap. 75 (2021) 6611–6623, https://doi.org/10.1007/s11696-021-01820-7.